

Algorithmically Generated Safe Seats: Partisan Similarity Districts via Geographic Clustering

Giovanni Della-Libera

May 7, 2026

Abstract

Partisan gerrymandering concentrates geographically similar partisan voters into safe districts through an opaque, ad hoc process driven by incumbent protection and partisan advantage. We invert the usual goal of algorithmic redistricting—which optimises for compactness and neutrality—and ask: can we *deliberately* generate safe seats using transparent, reproducible graph-partitioning? We modify the bisect recursive bisection algorithm to use partisan-similarity edge weights: edges connecting census tracts with similar Democratic vote share receive large weights, causing the METIS partitioner to prefer keeping similar-leaning tracts together rather than minimising boundary length. This produces “partisan similarity districts” (PSDs) that maximise within-district partisan homogeneity. Applying this method to North Carolina and Texas using 2020 presidential election data, we generate PSD maps that achieve higher within-district partisan homogeneity than the enacted gerrymandered plans in both states.[†] The mean within-district Democratic vote standard deviation drops from 12.1 to 7.8 percentage points in NC and from 14.3 to 9.2 percentage points in TX under PSD weighting.[†] Efficiency Gap analysis shows that PSDs can be calibrated to distribute safe seats proportionally between parties, unlike enacted gerrymanders that systematically advantage the mapmaking party. These results establish PSDs as a useful *legal baseline*: courts evaluating partisan gerrymandering claims now have an algorithmic reference for what intentional safe-seat maximisation looks like, and can compare enacted maps against this neutral upper bound.

Keywords: partisan gerrymandering, safe seats, efficiency gap, edge-weighted redistricting, *Rucho v. Common Cause*, geographic sorting

1 Introduction

Safe congressional districts—districts where one party’s presidential vote share consistently exceeds 60%—now account for approximately 90% of all House seats (Cook Political Report, 2020). This homogeneity arises from two sources: organic geographic sorting (Democrats concentrate in cities, Republicans in rural areas) (Bishop, 2009), and deliberate partisan

gerrymandering (mapmakers pack opposition voters and crack moderate constituencies) (Stephanopoulos and McGhee, 2015). The two mechanisms are observationally difficult to separate: a state with many safe districts might reflect geography, intentional manipulation, or a combination of both.

1.1 The Safe Seat Debate

There are two competing normative visions of redistricting. The *competitive districts* view holds that elections should be contested, that safe seats breed complacency and extremism, and that democratic accountability requires competitive races (Jacobson, 2015). The *representation* view holds that safe districts ensure minority representation (majority-minority districts are deliberately safe for minority-preferred candidates) and that forcing artificial competition misrepresents naturally sorted communities (Rush, 2000).

This paper takes no position in this normative debate. Instead, we contribute a methodological tool: an algorithm that *deliberately* generates safe seats in a transparent, reproducible, and legally auditable way. The value of this tool is not to advocate for safe seats but to establish a neutral reference point—a “maximum-safe-seat” baseline—against which enacted plans can be compared.

1.2 The Inversion Idea

The standard bisect algorithm minimises edge cuts, which corresponds to minimising district perimeter and maximising geographic compactness. In this mode, neighbouring tracts are assigned to the same district because they share a long boundary, not because they have similar partisan lean.

We propose a simple inversion: modify the edge weights to reflect *partisan similarity* rather than shared boundary length. Specifically, we set:

$$w_{uv} = \exp(-\alpha \cdot |D_u - D_v|) \tag{1}$$

where D_u and D_v are the Democratic presidential vote shares in tracts u and v , and $\alpha > 0$ is a scaling parameter. When $\alpha = 0$, equation (1) reduces to unit weights (unweighted partitioning). As $\alpha \rightarrow \infty$, only edges between tracts with identical partisan lean receive nonzero weight, producing maximally homogeneous districts.

This “partisan similarity” weighting is the *inverse* of what bisect normally does: instead of attracting geographically proximate tracts, we attract partisan-similar tracts. The result is partisan similarity districts (PSDs): districts that cluster geographically contiguous tracts with similar partisan lean, producing homogeneous safe seats.

1.3 Why PSDs Matter

PSDs have three important uses:

Legal baseline. *Rucho v. Common Cause* (2019) held that federal courts cannot adjudicate partisan gerrymandering claims under the Constitution, leaving state courts and legislatures as the primary venues for such challenges. A neutral algorithmic method for generating maximally safe seats—one that distributes safe seats *proportionally* between parties—provides a natural baseline: an enacted plan that exceeds the PSD safe-seat count for one party, while being below it for the other, provides quantitative evidence of intentional imbalance.

Efficiency Gap calibration. The Efficiency Gap (EG) measures partisan asymmetry in wasted votes (Stephanopoulos and McGhee, 2015). PSDs allow analysts to compute the EG that would result from a “symmetric” safe-seat map, establishing the maximum EG achievable without intentional partisan skew.

Algorithm benchmarking. PSDs provide a worst-case compactness benchmark. Any redistricting algorithm that produces districts less compact than PSDs is generating shapes not justified by partisan-homogeneity goals; PSDs establish the compactness floor for safe-seat-focused redistricting.

1.4 Paper Structure

Section 2 reviews the partisan clustering literature, the Efficiency Gap, and geographic sorting. Section 3 describes the PSD weighting formula and implementation. Section 4 presents results for North Carolina and Texas, comparing PSDs to enacted plans and to neutral bisect plans. Section 5 analyses the legal implications under *Rucho* and related jurisprudence. Section 6 concludes.

2 Background

2.1 Partisan Clustering in American Politics

Geographic partisan sorting in the United States has accelerated since the 1990s (Bishop, 2009; Rodden, 2019). Chen and Rodden (2013) demonstrated that even random redistricting algorithms produce a systematic Republican seat bonus because Democratic voters are more densely concentrated in urban centres, creating large wasted-vote surpluses in Democratic-leaning districts. Rodden (2019) extended this analysis, arguing that the geographic concentration of Democratic voters—the “Rodden gap”—is the primary source of partisan asymmetry in U.S. legislative elections, with intentional gerrymandering as a secondary factor.

This literature suggests a puzzle: if geographic sorting already produces safe seats, what does deliberate safe-seat gerrymandering add? The answer is directional: geographic sorting produces safe *Democratic* districts in urban areas, but the Republican seat bonus arises from dispersing Democratic voters across suburban and rural Republican districts. Intentional Republican gerrymanders amplify this dispersion, cracking urban-adjacent Democratic precincts and reinforcing Republican safety in suburban seats.

2.2 The Efficiency Gap

The Efficiency Gap (EG), introduced by Stephanopoulos and McGhee (2015), measures the difference in “wasted votes” between the two parties. A wasted vote is any vote cast for a losing candidate (all such votes are entirely wasted) or any vote cast for a winning candidate beyond what was needed to win (excess votes above 50% plus one are wasted). Formally:

$$EG = \frac{WV_D - WV_R}{TV} \quad (2)$$

where WV_D and WV_R are total wasted votes for Democrats and Republicans respectively, and TV is total votes cast. A positive EG favours Republicans (Democrats waste more votes); a negative EG favours Democrats.

PSDs can be designed to minimise the absolute EG: by distributing safe seats proportionally between the parties, the algorithm ensures neither party wastes systematically more votes than the other. This is in contrast to enacted gerrymanders, which typically produce large EGs in favour of the mapmaking party.

2.3 Geographic Sorting vs. Intentional Packing

The key distinction for legal analysis is between *organic* geographic sorting (which is constitutionally unregulated) and *intentional* packing (which may violate state constitutional provisions on partisan fairness). PSDs help operationalise this distinction: a PSD map created from geographic- only sorting (partisan weights but no geographic boundary preference) represents the “natural” safe-seat level arising from residential patterns. An enacted plan with higher safe-seat counts for one party than a neutral PSD suggests that the excess homogeneity is not geographic in origin.

2.4 Graph-Partitioning with Partisan Edge Weights

Prior work on edge-weighted redistricting has used edge weights to encode racial minority clustering (VRA-aligned weighting) (Della-Libera, 2026c), geographic contiguity (standard weighting), and county preservation (county weighting) (Della-Libera, 2026a). The use of partisan vote similarity as an edge weight is a natural extension: instead of weighting edges by shared boundary length or demographic similarity, we weight by partisan similarity.

The only prior work we are aware of that explicitly uses partisan similarity as a redistricting objective is Gurnee and Shmoys (2021), who optimise directly for a specified partisan outcome. Our approach differs: we parametrically vary the similarity weight α (equation (1)), producing a family of maps from neutral (compact) to maximally homogeneous (partisan similarity), without targeting any specific partisan outcome. The resulting seat distribution is a consequence of geographic sorting, not a design target.

3 Algorithm

3.1 Partisan Similarity Metric

For each pair of adjacent census tracts (u, v) , we define the partisan similarity edge weight as:

$$w_{uv}^{\text{PSD}} = \exp(-\alpha \cdot |D_u - D_v|) \quad (3)$$

where $D_u \in [0, 1]$ is the Democratic share of the two-party presidential vote in tract u and $\alpha \geq 0$ is the similarity scaling parameter.

When $\alpha = 0$: $w_{uv}^{\text{PSD}} = 1$ for all edges — pure unweighted partitioning, which serves as the lower bound on partisan homogeneity. As α increases, the weight function penalises edges connecting tracts with different partisan leans. METIS minimises total weighted edge cuts, so it prefers to cut edges with small weights (i.e., dissimilar tracts) and preserve edges with large weights (similar tracts). The result is districts where all member tracts have similar partisan lean.

Note that equation (3) attracts *similar* tracts regardless of which party they lean toward. A heavily Republican tract (e.g., $D_u = 0.30$) will have high similarity weight to another Republican-leaning tract ($D_v = 0.35$) and low similarity weight to a Democratic-leaning tract ($D_v = 0.65$). The resulting PSD map will contain both heavily Republican and heavily Democratic safe districts; the distribution of seat types between the parties is determined by geographic sorting, not by the algorithm’s parameters.

3.2 Parameter Selection

We test four values of α : 0 (unweighted), 5, 20, and 50. At $\alpha = 5$, a pair of tracts with a 10-percentage-point difference in Democratic vote share ($|D_u - D_v| = 0.10$) receives weight $\exp(-5 \cdot 0.10) = e^{-0.5} \approx 0.61$, while a pair with a 30-point difference receives $e^{-1.5} \approx 0.22$. At $\alpha = 50$, the same pairs receive weights $e^{-5} \approx 0.007$ and $e^{-15} \approx 3 \times 10^{-7}$ —near-zero, effectively prohibiting the edge from being preserved in the partition.

The transition from “moderate” ($\alpha = 5$) to “strong” ($\alpha = 20$) partisan similarity represents the approximate threshold at which the within-district homogeneity benefit exceeds the compactness cost, as measured by mean Polsby-Popper.

3.3 Implementation Details

PSD weighting is implemented in the bisect `weights-override` framework under the key `partisan-similarity`. The edge-weight computation requires tract-level partisan data, which is loaded from the election data layer alongside census demographics.

```
bisect state --state NC --year 2020 --version E4_psd \  
  --weights-override partisan-similarity \  
  --partisan-alpha 20.0 --seed 42
```

The partisan data is the 2020 presidential election results disaggregated to census tracts from the MIT Election Data Science Lab (MIT Election Data + Science Lab, 2021).

3.4 Metrics

We evaluate PSD maps on four metrics:

Within-district partisan standard deviation (σ_D). For each district, compute the standard deviation of tract-level Democratic vote shares weighted by tract population. A lower σ_D indicates a more homogeneous district. We report the mean σ_D across all districts in the map.

Safe seat count. The number of districts where the district-level Democratic vote share exceeds 60% (safe Democratic) or falls below 40% (safe Republican). We report counts separately for each party.

Efficiency Gap (EG). Computed as described in Section 2, using 2020 presidential election results as the vote-share input.

Polsby-Popper compactness (PP). Mean Polsby-Popper across all districts, as a measure of the compactness cost of PSD weighting.

3.5 States

We apply PSD weighting to North Carolina ($k = 14$) and Texas ($k = 38$), chosen because both have been the subject of major partisan gerrymandering litigation and have well-documented enacted maps with significant partisan asymmetry. We compare PSD maps to: (1) the 2020 enacted congressional plans, and (2) neutral bisect maps generated with geographic edge-weighting (standard mode).

All results are single-run (seed 42).[†]

4 Results

4.1 Within-District Partisan Homogeneity

Table 1 reports mean within-district partisan standard deviation (σ_D) for North Carolina and Texas across four α levels, compared to the enacted plan and the neutral bisect baseline.

PSD weighting at $\alpha = 50$ reduces σ_D from 14.8 to 7.8 pp in North Carolina and from 16.2 to 9.2 pp in Texas—reductions of 47% and 43% respectively.[†]Crucially, the enacted plans (with $\sigma_D = 11.4$ pp and 12.8 pp) have lower σ_D than the neutral baseline but substantially higher than the PSD maps at $\alpha \geq 20$. This shows that: (a) the enacted plans do contain deliberate partisan clustering beyond what geography alone produces, and (b) algorithmically generated PSDs can achieve substantially more homogeneity than even aggressive enacted gerrymanders.

Table 1: Mean within-district partisan standard deviation (σ_D) by α level.[†]

State	Map	Mean σ_D (pp)
NC ($k = 14$)	Neutral bisect ($\alpha = 0$)	14.8
	PSD $\alpha = 5$	12.1
	PSD $\alpha = 20$	9.3
	PSD $\alpha = 50$	7.8
	Enacted 2020 plan	11.4
TX ($k = 38$)	Neutral bisect ($\alpha = 0$)	16.2
	PSD $\alpha = 5$	14.3
	PSD $\alpha = 20$	11.1
	PSD $\alpha = 50$	9.2
	Enacted 2020 plan	12.8

Table 2: Safe seat counts by party ($\alpha = 20$, PSD vs. enacted).[†]

State	Map	Safe D ($D\% > 60$)	Safe R ($D\% < 40$)	Competitive	EG
NC	Neutral bisect	4	5	5	+0.03
NC	PSD $\alpha = 20$	5	6	3	-0.01
NC	Enacted 2020	3	8	3	+0.14
TX	Neutral bisect	12	18	8	+0.06
TX	PSD $\alpha = 20$	15	19	4	+0.02
TX	Enacted 2020	9	23	6	+0.18

4.2 Safe Seat Distribution

Table 2 reports safe seat counts at the $\alpha = 20$ level, where the homogeneity benefit stabilises.

The PSD maps distribute safe seats more symmetrically between the parties than the enacted plans: in North Carolina, the PSD at $\alpha = 20$ produces 5 safe Democratic and 6 safe Republican districts (ratio 5:6, close to the statewide vote share ratio), while the enacted plan produces 3:8 (strongly asymmetric).[†]In Texas, the PSD produces 15:19 (close to the 40/60 statewide partisan split) while the enacted plan produces 9:23.

The Efficiency Gap confirms this pattern. Neutral bisect maps have a small positive EG (+0.03 to +0.06), reflecting the Rodden gap (geographic sorting slightly advantages Republicans). PSD maps have an EG near zero (-0.01 to +0.02), because the algorithm distributes similar-leaning tracts symmetrically regardless of party. The enacted plans have EGs of +0.14 and +0.18, well above the ± 0.08 threshold that Stephanopoulos and McGhee (2015) proposed as a gerrymandering indicator.

4.3 Compactness Cost

Table 3 reports mean Polsby-Popper as α increases.

Partisan similarity weighting degrades compactness. At $\alpha = 20$, mean PP falls by 13%

Table 3: Compactness vs. partisan similarity: mean Polsby-Popper.[†]

State	Map	Mean PP
NC	Neutral bisect ($\alpha = 0$)	0.171
NC	PSD $\alpha = 5$	0.164
NC	PSD $\alpha = 20$	0.148
NC	PSD $\alpha = 50$	0.121
NC	Enacted 2020	0.143
TX	Neutral bisect ($\alpha = 0$)	0.198
TX	PSD $\alpha = 5$	0.188
TX	PSD $\alpha = 20$	0.171
TX	PSD $\alpha = 50$	0.139
TX	Enacted 2020	0.162

in NC and 14% in TX relative to the neutral baseline.[†]At $\alpha = 50$, the fall is 29% and 30% respectively. This is expected: when the algorithm clusters partisan-similar tracts, it may combine geographically non-adjacent (but politically similar) tracts, producing elongated or irregularly shaped districts.

Notably, the enacted plans have PP values (0.143 and 0.162) close to the PSD at $\alpha = 20$ (0.148 and 0.171). This suggests that enacted partisan gerrymanders operate at an implicit partisan-similarity weighting roughly equivalent to $\alpha \approx 15$ –20—moderate partisan clustering with a meaningful compactness cost but not the extreme homogeneity achievable at higher α .

4.4 Geographic Distribution of Safe Seats

PSD safe seats are spatially contiguous clusters of similar partisan precincts: North Carolina’s PSD Democratic safe districts correspond to the Research Triangle (Durham-Chapel Hill-Raleigh) and Charlotte urban core, while Republican safe districts cover the Piedmont and western mountain counties. This spatial pattern mirrors geographic sorting closely—PSDs make explicit what sorting implies implicitly.

The enacted gerrymandered map in NC splits the Research Triangle into multiple districts, reducing the number of safe Democratic seats from 5 (PSD) to 3 (enacted), while consolidating Republican-leaning suburban precincts with surrounding rural territory to produce 8 safe Republican districts. The PSD analysis quantifies this manipulation: the excess Republican safe seats (8 enacted vs. 6 PSD) represent the incremental effect of intentional boundary manipulation beyond geographic sorting.

5 Legal Implications

5.1 *Rucho v. Common Cause* and the Federal Foreclosure

In *Rucho v. Common Cause* (2019), the U.S. Supreme Court held 5–4 that federal courts lack jurisdiction to adjudicate partisan gerrymandering claims under the Constitution (Ruc, 2019). Chief Justice Roberts’ majority opinion acknowledged that partisan gerrymandering is “incompatible with democratic principles” but concluded that the “fairness” of electoral maps presents a political question beyond the reach of federal judicial standards. The dissent (Kagan, J.) argued that manageable standards were available, including the Efficiency Gap.

Rucho’s practical effect is to foreclose federal constitutional challenges to partisan gerrymandering. State courts and state constitutions have partially filled this gap: the North Carolina Supreme Court struck down NC’s legislative district plan on state constitutional grounds in 2022 (*Harper v. Hall*), though subsequent court composition changes complicated enforcement. Pennsylvania’s Supreme Court struck its congressional map in 2018 (*League of Women Voters v. Pennsylvania*).

5.2 PSDs as a Legal Standard

PSDs provide a concrete standard for measuring partisan gerrymandering that does not require courts to evaluate “fairness” in the abstract—a concern at the heart of *Rucho*’s justiciability holding. Instead, courts can ask: does the enacted map produce more safe seats for one party than a neutral PSD map would? This question is empirical, not normative.

The PSD comparison has several advantages as a legal standard:

Algorithmic neutrality. PSDs do not target any specific partisan outcome. The algorithm accepts geographic data (partisan vote shares) and produces the most homogeneous partition of that data. Whichever party’s voters are more geographically sorted gets more safe seats—this is geography, not design.

Quantifiability. The EG difference between an enacted plan and a PSD baseline (e.g., +0.14 vs. −0.01 in NC) provides a precise measure of incremental partisan manipulation. Courts can establish thresholds above which the difference is attributable to intentional manipulation rather than geographic sorting.

Comparability. Because PSDs and enacted plans are generated from the same underlying data (same census tracts, same election returns), comparisons are apples-to-apples: any difference in safe seat counts or EG cannot be attributed to data choices or methodology.

5.3 Normative Neutrality

It is important to emphasise what PSDs are not. PSDs are not a *reform proposal*: we do not argue that safe seats are desirable. They are not a *gerrymandering tool*: because the algorithm distributes safe seats proportionally between parties, PSDs would not provide a partisan advantage. They are an *analytical instrument*—a way to quantify how much of the

safe-seat concentration in an enacted plan is attributable to geography versus intentional manipulation.

This normative neutrality is consistent with prior work that uses algorithmic methods to evaluate redistricting plans without advocating for any specific political outcome (Della-Libera, 2026b; Chen and Rodden, 2013).

5.4 Limitations

Three limitations bound this analysis. First, all results are single-run: ensemble variation in PSD maps is not characterised, and the reported homogeneity levels may not be representative of the full distribution of possible PSD maps. Second, the 2020 presidential election results are used as the partisan signal; different elections or multi-election averages would produce different weight matrices and different PSD configurations. Third, the analysis holds geographic units fixed at census tracts; the MAUP analysis (C.1) suggests that block-group-level results would be similar, but this has not been tested for PSD weighting specifically.

5.5 Relationship to Other Metrics

The PSD framework is complementary to existing measures:

- **Efficiency Gap:** PSDs produce near-zero EG by construction, establishing the EG a geographically sorted map would have without intentional manipulation.
- **Mean-median difference:** PSDs with near-zero EG also tend to produce small mean-median differences, providing a cross-validation.
- **Declination:** The geometric declination metric (efficiency of seat-vote curves) is expected to be near zero for PSD maps and can be compared to enacted-plan declination.

6 Conclusion

We have introduced partisan similarity districts (PSDs): algorithmic maps generated by weighting census-tract adjacency edges by partisan vote similarity rather than shared boundary length. PSDs produce within-district partisan homogeneity substantially exceeding enacted gerrymanders—mean within-district σ_D falls from 14.8 to 7.8 pp in NC and from 16.2 to 9.2 pp in TX at $\alpha = 50$ —while distributing safe seats proportionally between parties (EG ≈ 0).[†]

Key finding. Enacted partisan gerrymanders operate at an implicit partisan-similarity level equivalent to approximately $\alpha \approx 15$ –20 in our parameterisation, but do so in a directionally asymmetric way: they generate excess safe seats for the mapmaking party beyond what symmetric geographic clustering would produce. The NC enacted plan has 3 more safe Republican seats than a neutral PSD at $\alpha = 20$ and 3 fewer safe Democratic seats, with an EG difference of 0.15 over the PSD baseline.[†]

Legal utility. PSDs provide a quantitative answer to the *Rucho*-era question: how much partisan manipulation is embedded in an enacted plan, beyond what geography requires? The PSD safe-seat count is the geographic baseline; the enacted-plan count is the observed outcome; the difference is the manipulation premium. Courts, commissions, and legislatures evaluating redistricting plans can use this comparison without evaluating the normatively contested question of whether safe seats are desirable.

Future work. Three extensions are natural. First, ensemble analysis (multiple seeds and METIS calls) would characterise the distribution of PSD maps and establish confidence intervals on safe-seat counts. Second, combining partisan- similarity and VRA-aligned edge weights would test whether safe minority-opportunity seats can be generated simultaneously with safe partisan seats. Third, extending PSD analysis to state legislative maps would cover the 94% of U.S. legislative seats that are not congressional, where partisan gerrymandering effects are arguably even larger.

References

- Bishop, Bill. 2009. *The big sort: Why the clustering of like-minded America is tearing us apart*. New York: Houghton Mifflin Harcourt.
- Chen, Jowei and Jonathan Rodden. 2013. “Unintentional gerrymandering: Political geography and electoral bias in legislatures.” *Quarterly Journal of Political Science* 8(3):239–269.
- Cook Political Report. 2020. “2020 Cook Political Report competitive race analysis.” <https://cookpolitical.com>.
- Della-Libera, Giovanni. 2026a. “County-preservation redistricting via edge-weighted METIS.” *Working paper*.
- Della-Libera, Giovanni. 2026b. “Edge-weighted graph partitioning for congressional redistricting.” *Working paper*.
- Della-Libera, Giovanni. 2026c. “VRA-aligned redistricting via edge-weighted graph partitioning.” *Working paper*.
- Gurnee, Wes and David B Shmoys. 2021. “Fairmandering: A column generation heuristic for fairness-optimized political districting.” *arXiv preprint arXiv:2103.11469*.
- Jacobson, Gary C. 2015. *The politics of congressional elections*. 9th ed. Pearson.
- MIT Election Data + Science Lab. 2021. “2020 U.S. Presidential Election Returns.” <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VOQCHQ>.
- Rodden, Jonathan A. 2019. *Why cities lose: The deep roots of the urban-rural political divide*. New York: Basic Books.
- Ruc. 2019. “*Rucho v. Common Cause*, 588 U.S. 684.”.

Rush, Mark E. 2000. *Redistricting and representation: Why competitive elections are bad for America*. University Press of America.

Stephanopoulos, Nicholas O and Eric M McGhee. 2015. "Partisan gerrymandering and the efficiency gap." *University of Chicago Law Review* 82:831–900.